

# Typhoons: an example of stepwise selection

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**Credits:** I have adapted an example from Richard Chandler.

## Introduction

We shall consider tropical storms in the North-West Pacific Ocean, for each year between 1959 and 2000. It is well known that the state of El Nino is related to tropical storm activity in the North-West Pacific the following year.

Data in the dataset `typhoons.txt`

- `Typhoons` are events with a maximum windspeed over 64 knots;
- `N3.mxx` Nino 3 anomalies for each month (`xx`) from the previous year,

## Reading the dataset

```
alldata<-read.table("typhoons.txt",header = TRUE)
```

## Exploratory analysis: summary statistics and visualizing data

```
table(alldata$Typhoons)
```

```
##  
##  9 11 12 13 14 15 16 17 18 19 20 21 22 23 24 26  
##  1  2  3  2  5  2  3  2  1  4  6  5  1  3  1  1
```

```
summary(alldata$Typhoons)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##      9.00  14.00   18.50   17.50   20.75   26.00
```

```
par(mfrow=c(1,3),pty='s')
```

```
for (i in 1:12){
```

```
  plot(alldata$Typhoons,alldata[,i+2],ylab="Number of typhoons", xlab="Nino (degrees C)",cex=1.3,pch=20)
```

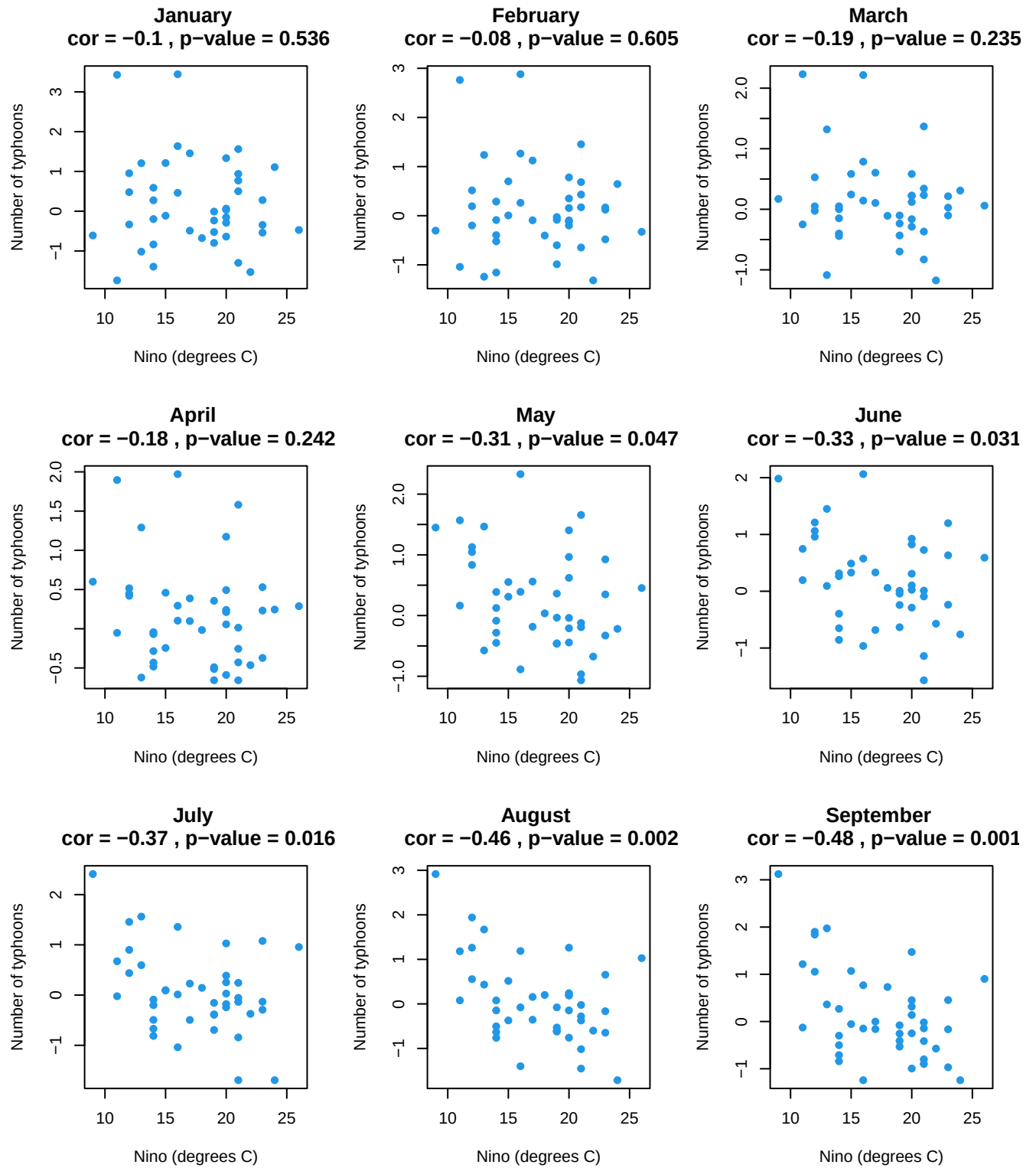
```
  ct<-cor.test(alldata$Typhoons,alldata[,i+2])
```

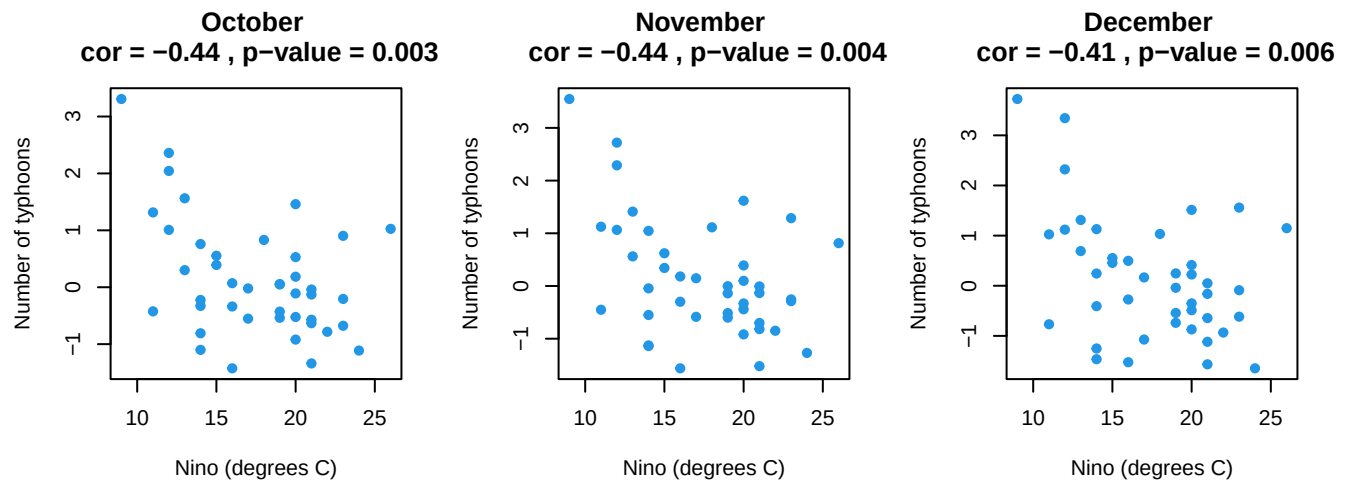
```
  r<-ct$estimate
```

```
  p.value<-ct$p.value
```

```
  title(main=paste(month.name[i],"\n cor =",round(r,2),", p-value =", round(p.value,3)))
```

```
}
```

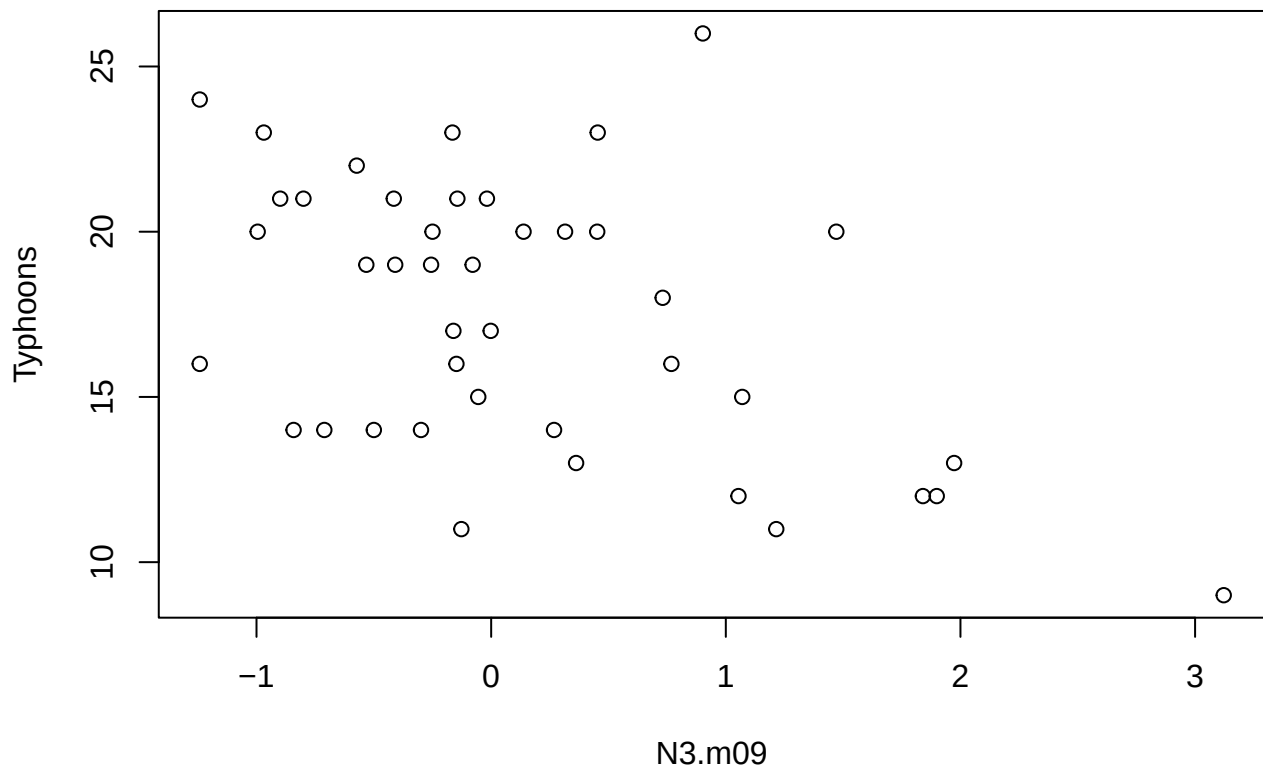




```
round(cor(alldata)[2,-1],4)
```

```
## Typhoons    N3.m01    N3.m02    N3.m03    N3.m04    N3.m05    N3.m06    N3.m07
## 1.0000 -0.0982 -0.0822 -0.1874 -0.1847 -0.3082 -0.3325 -0.3695
## N3.m08    N3.m09    N3.m10    N3.m11    N3.m12
## -0.4632 -0.4813 -0.4422 -0.4373 -0.4140
```

```
plot(Typhoons~N3.m09,data=alldata)
```

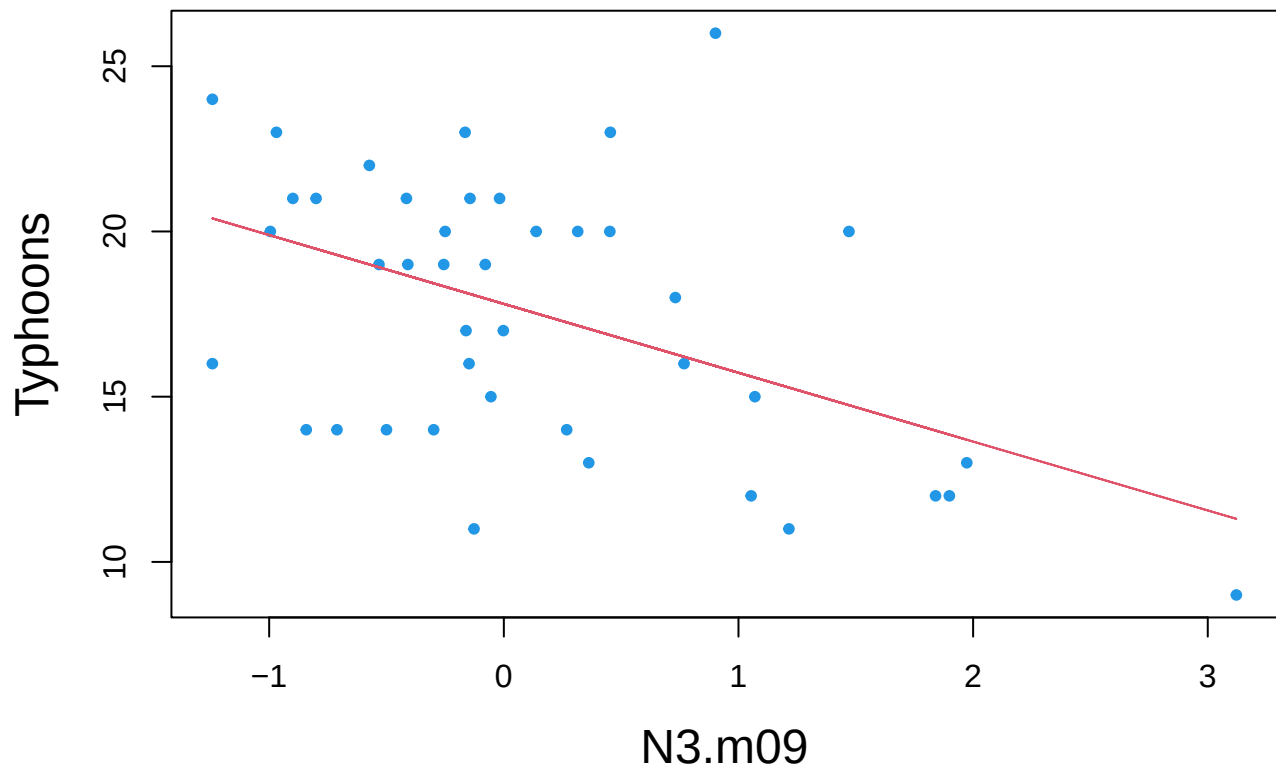


Should a linear model on the original data correct ?

```
fit<-lm(Typhoons~N3.m09,data=alldata)
```

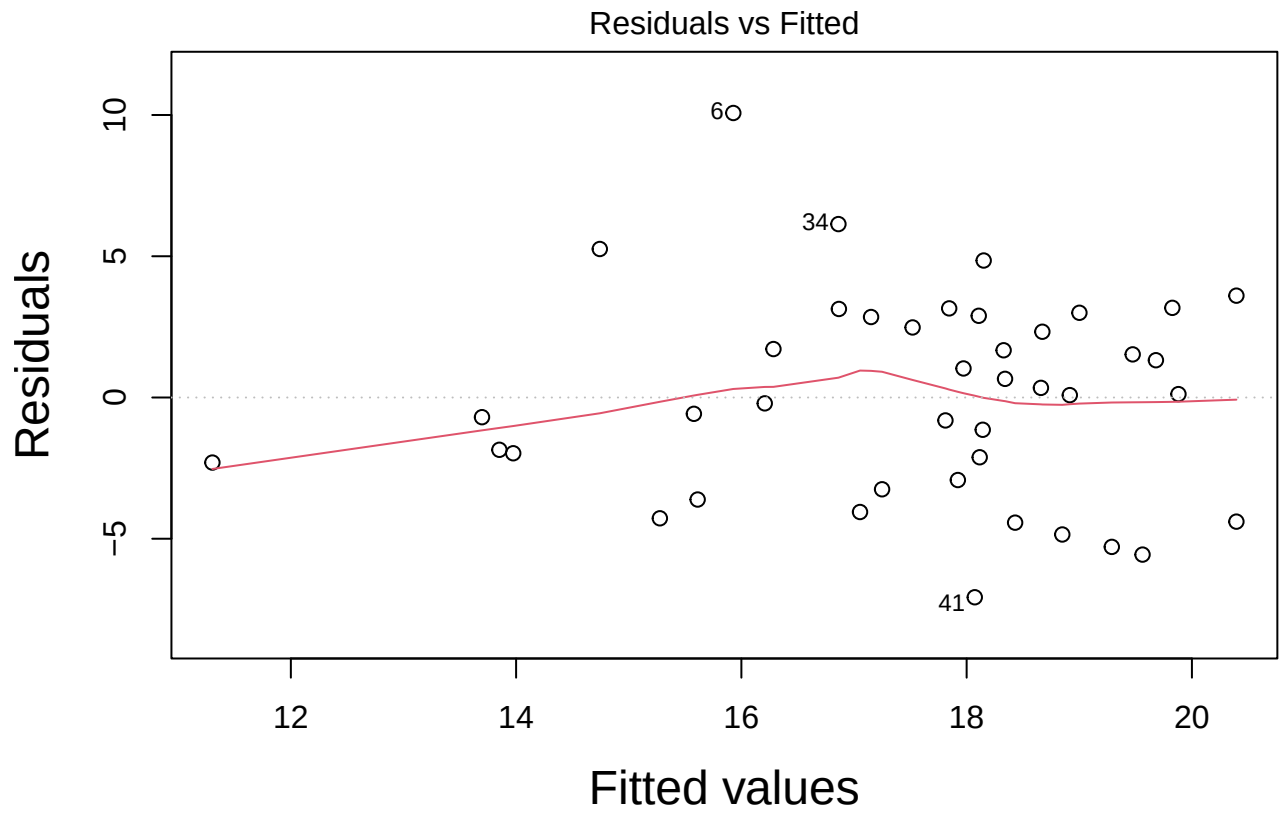
Fitted model

```
plot(Typhoons~N3.m09,data=alldata,col=4,pch=20,cex.lab=1.5)  
lines(alldata$N3.m09,fitted(fit),col=2)
```

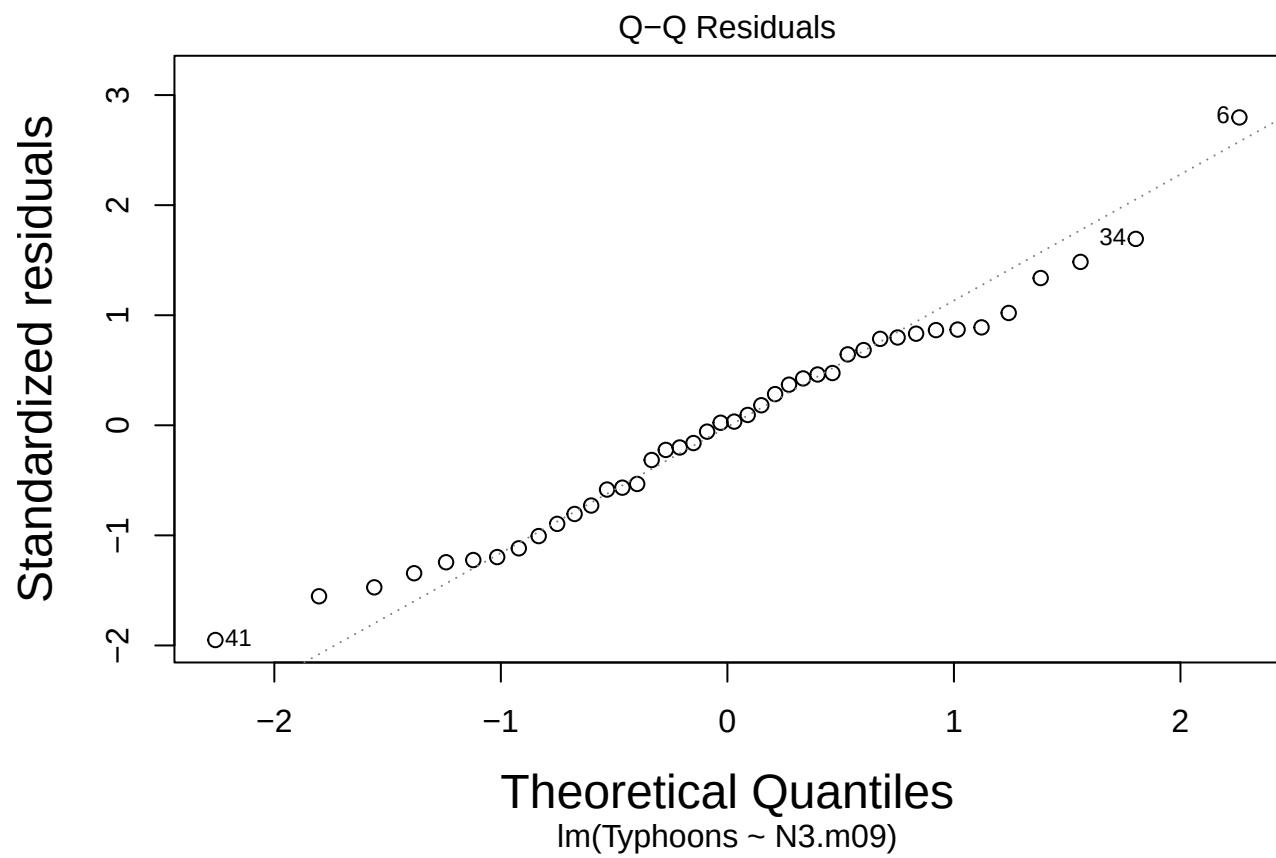


Analysis of residuals

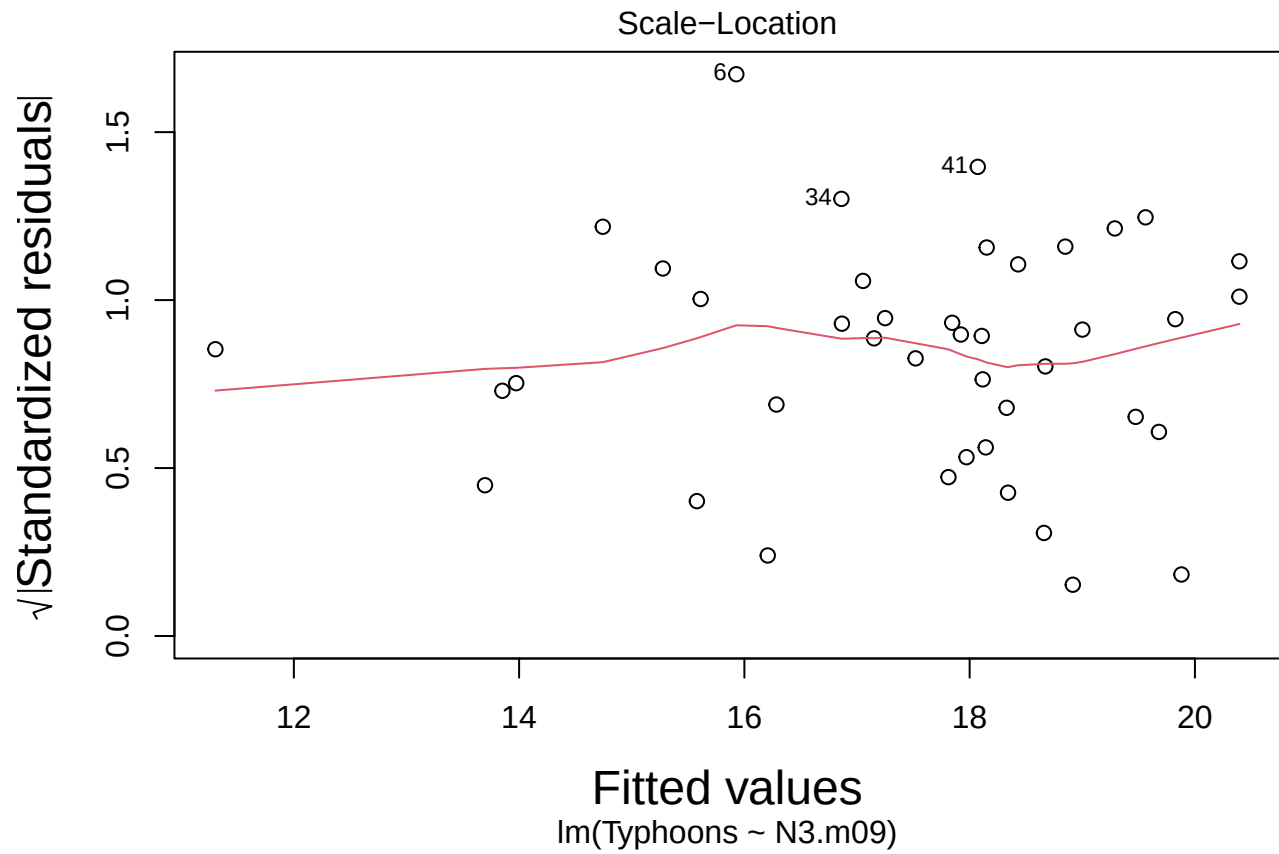
```
plot(fit,which = 1,sub.caption = "",cex.lab=1.5)
```



```
plot(fit,which = 2,cex.lab=1.5)
```



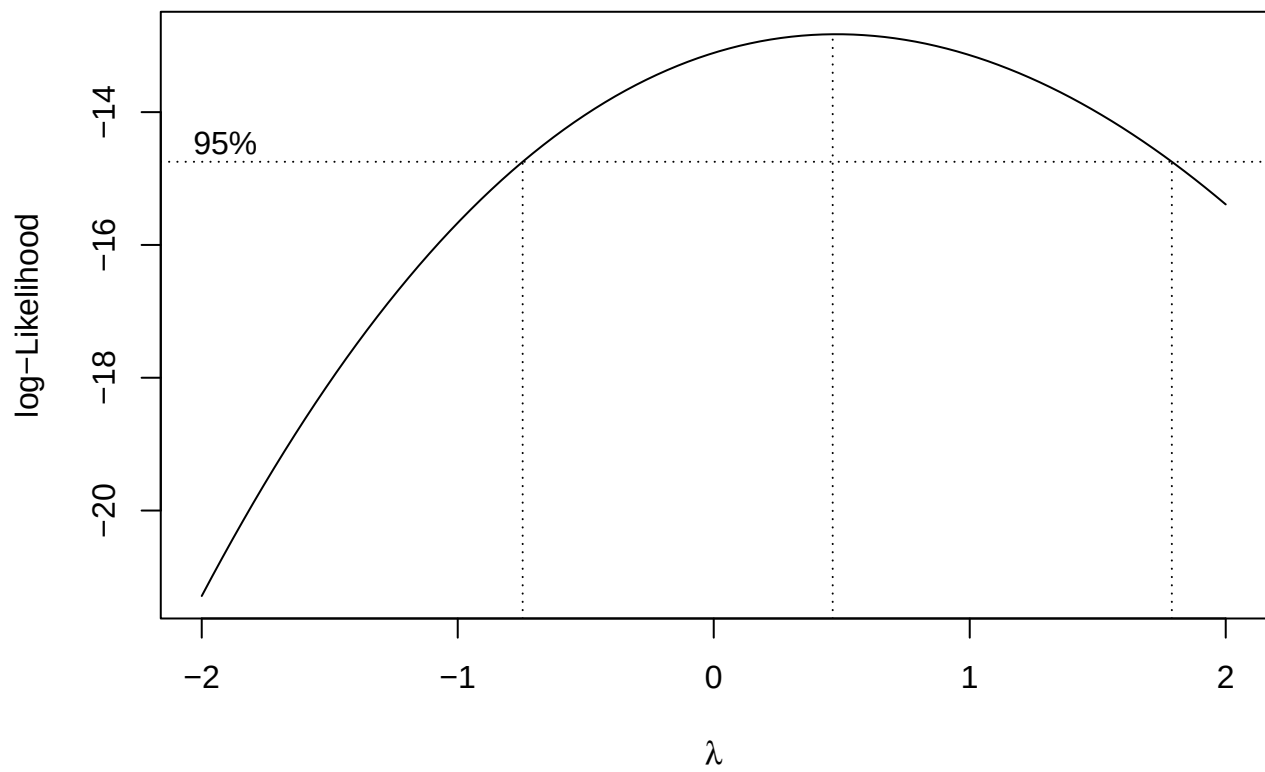
```
plot(fit,which = 3,cex.lab=1.5)
```



Variability of residuals depends on the regressor.

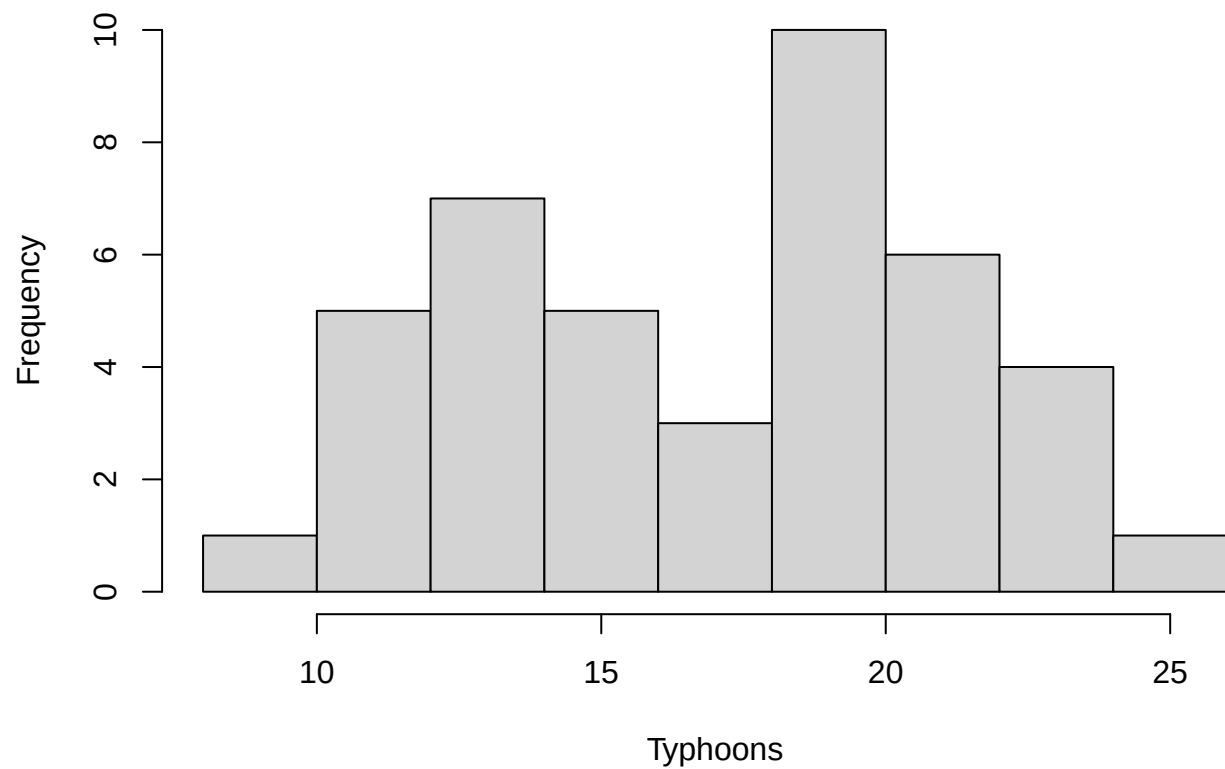
## Transforming the dependent variable ?

```
library(MASS)
boxcox(Typhoons ~ N3.m09, data=alldata)
```



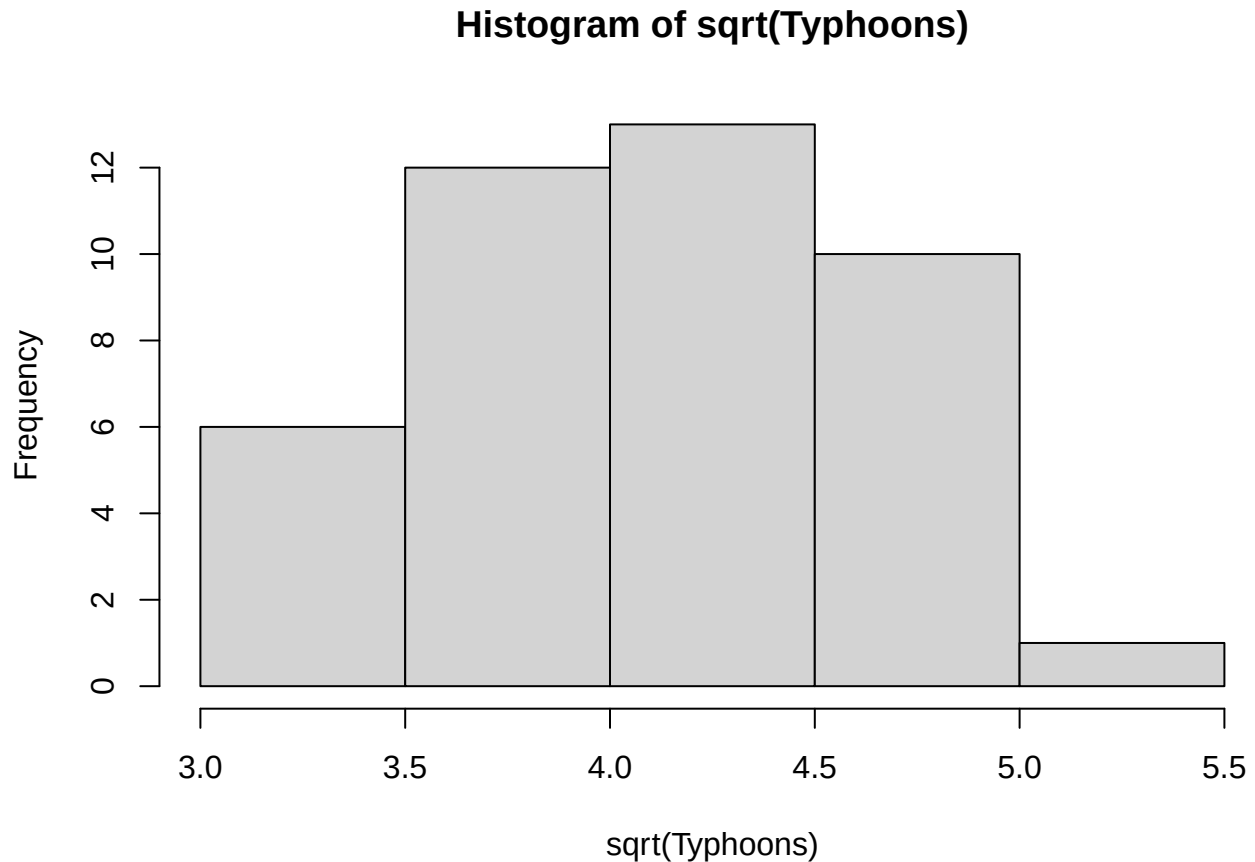
```
attach(alldata)
hist(Typhoons)
```

**Histogram of Typhoons**





```
hist(sqrt(Typhoons))
```



```
Typhoons.sqrt<-sqrt(Typhoons)
```

## A first linear model

The month (September) with the higher absolute correlation is chosen as the predictor variable.

```
fit.2<-lm(Typhoons.sqrt~N3.m09)
summary(fit.2)
```

```
##
## Call:
## lm(formula = Typhoons.sqrt ~ N3.m09)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.91025 -0.35067  0.01824  0.33653  1.14795
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.19283    0.06907  60.703  < 2e-16 ***
## N3.m09        -0.26803    0.07222  -3.711  0.000628 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.4423 on 40 degrees of freedom
## Multiple R-squared:  0.2561, Adjusted R-squared:  0.2375
## F-statistic: 13.77 on 1 and 40 DF,  p-value: 0.0006277
```

We look at the IC

```
AIC(fit.2)
```

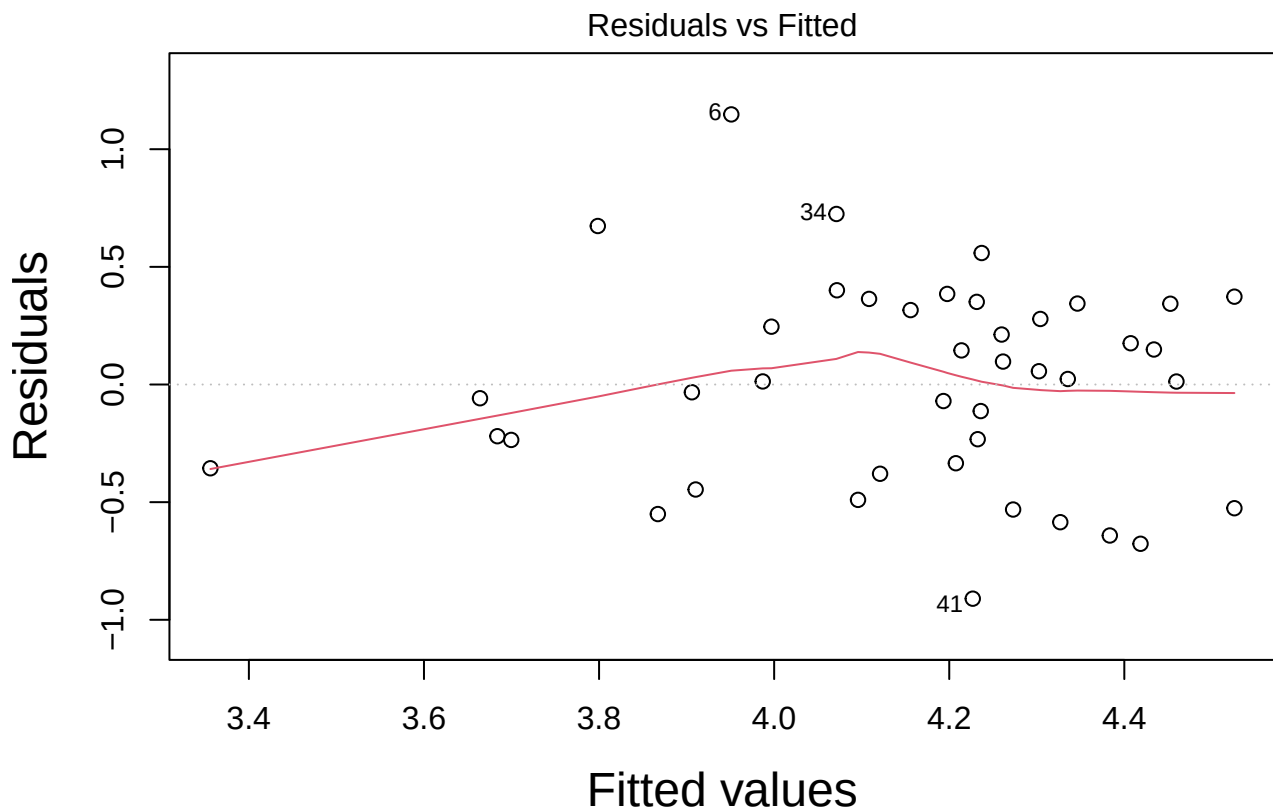
```
## [1] 54.61027
```

```
BIC(fit.2)
```

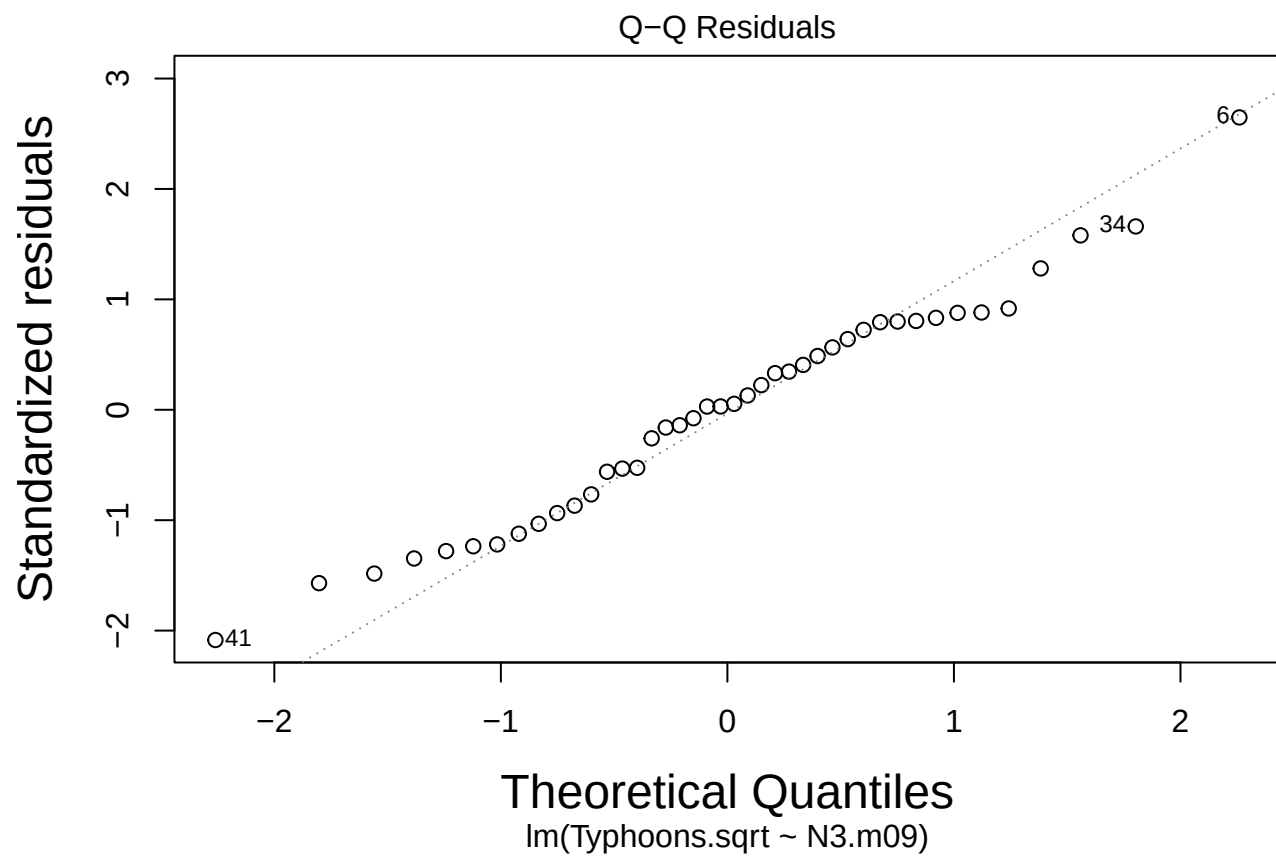
```
## [1] 59.82328
```

Analysis of residuals

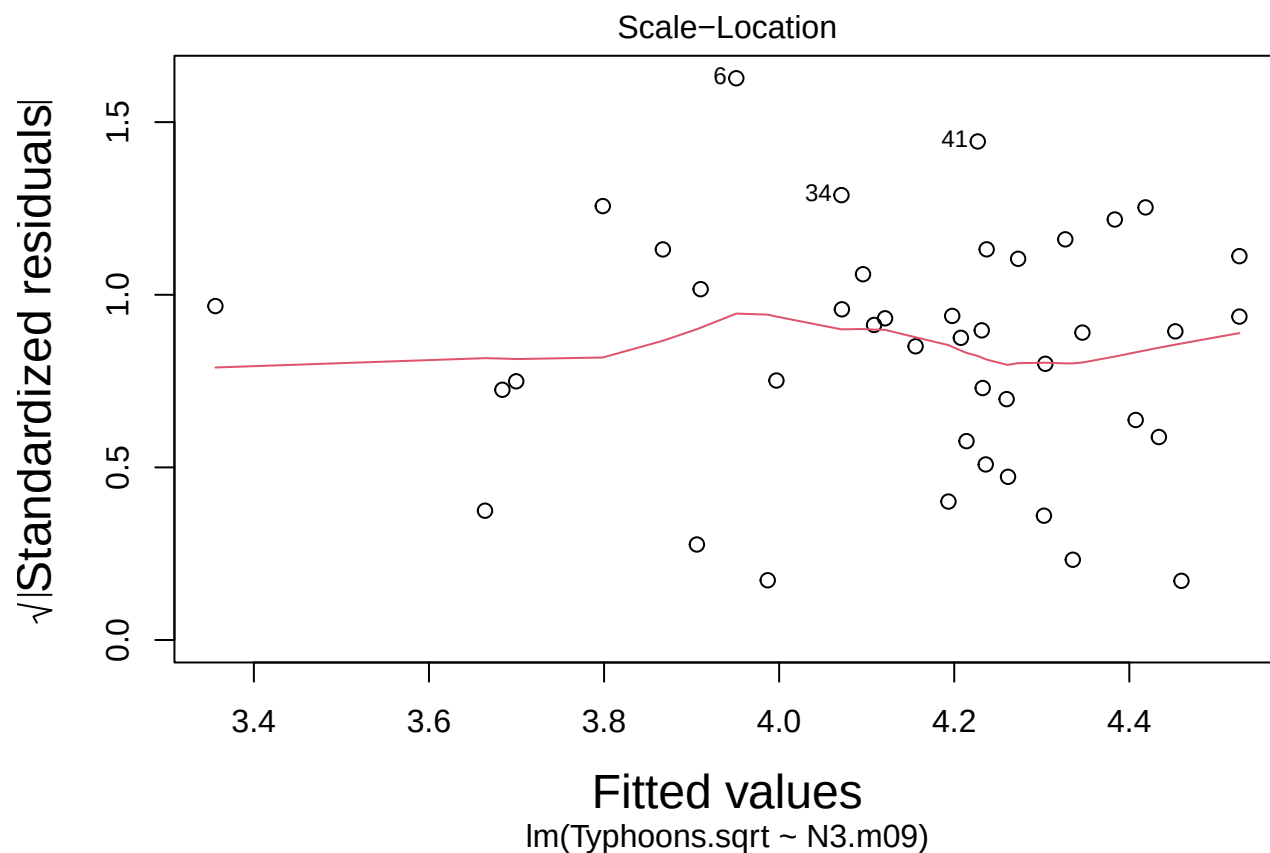
```
plot(fit.2,which = 1,sub.caption = "",cex.lab=1.5)
```



```
plot(fit.2,which = 2,cex.lab=1.5)
```



```
plot(fit.2,which = 3,cex.lab=1.5)
```



### Technical note

The log-likelihood and hence the AIC/BIC is only defined up to an additive constant. Different constants have conventionally been used for different purposes and so `extractAIC` and `AIC` may give different values

```
k <- 2
AIC(fit.2, k = k)
```

```
## [1] 54.61027
```

```
extractAIC(fit.2, k = k)[2]
```

```
## [1] -66.58056
```

```
BIC(fit.2)
```

```
## [1] 59.82328
```

is equivalent to

```
k <- log(length(Typhoons.sqrt))
AIC(fit.2, k = k)
```

```
## [1] 59.82328
```

```
extractAIC(fit.2, k = k)[2]
```

```
## [1] -63.10522
```

The additive constant for AIC

```
k <- 2
AIC(fit.2, k = k) - extractAIC(fit.2, k = k)[2]
```

```
## [1] 121.1908
```

The additive constant for BIC

```
k <- log(length(Typhoons.sqrt))
AIC(fit.2, k = k) - extractAIC(fit.2, k = k)[2]
```

```
## [1] 122.9285
```

## Stepwise selection

### Forward selection by means of AIC

The full model

```
fit.full <- lm(Typhoons.sqrt ~ ., data = alldata[, -(1:2)])
summary(fit.full)
```

```
##
## Call:
## lm(formula = Typhoons.sqrt ~ ., data = alldata[, -(1:2)])
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.70583 -0.18600 -0.06179  0.15262  0.97250
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.17737    0.08364  49.942  <2e-16 ***
## N3.m01        -0.37370    0.30553  -1.223   0.2311
## N3.m02         0.51275    0.51057   1.004   0.3236
## N3.m03        -0.44862    0.42523  -1.055   0.3001
## N3.m04         0.37219    0.33791   1.101   0.2798
## N3.m05         0.01669    0.37330   0.045   0.9646
## N3.m06        -0.21176    0.40585  -0.522   0.6058
## N3.m07         0.87072    0.48045   1.812   0.0803 .
## N3.m08        -0.98856    0.50777  -1.947   0.0613 .
## N3.m09         0.10338    0.48285   0.214   0.8320
## N3.m10         0.20453    0.57883   0.353   0.7264
## N3.m11        -0.49327    0.60754  -0.812   0.4235
## N3.m12         0.22542    0.40542   0.556   0.5825
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4491 on 29 degrees of freedom
## Multiple R-squared:  0.4438, Adjusted R-squared:  0.2137
## F-statistic: 1.928 on 12 and 29 DF,  p-value: 0.07293
extractAIC(fit.full)[2]
```

```
## [1] -56.79302
```

The null model i.e. the model with the only constant

$$Y_i = \beta_0 + \varepsilon_i$$

```
fit.null<-lm(Typhoons.sqrt~1)
summary(fit.null)

##
## Call:
## lm(formula = Typhoons.sqrt ~ 1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.1533 -0.4116  0.1475  0.4017  0.9458
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.15326    0.07815   53.14  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5065 on 41 degrees of freedom
extractAIC(fit.null)[2]

## [1] -56.15334
k <- 2
step(fit.null,scope=list(lower=fit.null,upper=fit.full), direction="forward",k=k,trace=1)

## Start:  AIC=-56.15
## Typhoons.sqrt ~ 1
##
##           Df Sum of Sq  RSS    AIC
## + N3.m09   1   2.69389  7.8239 -66.581
## + N3.m08   1   2.51842  7.9994 -65.649
## + N3.m10   1   2.32396  8.1938 -64.640
## + N3.m11   1   2.26156  8.2562 -64.322
## + N3.m12   1   2.02808  8.4897 -63.150
## + N3.m07   1   1.62836  8.8894 -61.218
## + N3.m06   1   1.30494  9.2128 -59.717
## + N3.m05   1   1.09305  9.4247 -58.762
## <none>                 10.5178 -56.153
## + N3.m04   1   0.40213 10.1156 -55.791
## + N3.m03   1   0.37546 10.1423 -55.680
## + N3.m01   1   0.08759 10.4302 -54.505
## + N3.m02   1   0.06257 10.4552 -54.404
##
## Step:  AIC=-66.58
## Typhoons.sqrt ~ N3.m09
##
##           Df Sum of Sq  RSS    AIC
## <none>                 7.8239 -66.581
## + N3.m01   1   0.181211  7.6427 -65.565
## + N3.m07   1   0.162996  7.6609 -65.465
## + N3.m03   1   0.115976  7.7079 -65.208
## + N3.m06   1   0.075114  7.7488 -64.986
```

```
## + N3.m02  1  0.071314  7.7526 -64.965
## + N3.m10  1  0.035293  7.7886 -64.770
## + N3.m12  1  0.016305  7.8076 -64.668
## + N3.m08  1  0.004905  7.8190 -64.607
## + N3.m05  1  0.001698  7.8222 -64.590
## + N3.m11  1  0.001294  7.8226 -64.588
## + N3.m04  1  0.000271  7.8236 -64.582

##
## Call:
## lm(formula = Typhoons.sqrt ~ N3.m09)
##
## Coefficients:
## (Intercept)      N3.m09
##          4.193      -0.268
```

The first argument is an object representing a model of class `lm`. This is used as the initial model in the stepwise search.

The set of models searched is determined by the `scope` argument.

$k$  the multiple of the number of regression parameters used for the penalty.

- $k = 2$ : AIC
- $k = \log(n)$ : BIC ( $n$  number of observations) `## Backward selection by means of AIC`

```
step(fit.full,scope=list(lower=fit.null,upper=fit.full),
     direction="backward",k=k,trace=1)
```

```
## Start:  AIC=-56.79
## Typhoons.sqrt ~ N3.m01 + N3.m02 + N3.m03 + N3.m04 + N3.m05 +
##      N3.m06 + N3.m07 + N3.m08 + N3.m09 + N3.m10 + N3.m11 + N3.m12
##
##      Df Sum of Sq  RSS    AIC
## - N3.m05  1   0.00040 5.8502 -58.790
## - N3.m09  1   0.00925 5.8590 -58.727
## - N3.m10  1   0.02519 5.8750 -58.613
## - N3.m06  1   0.05491 5.9047 -58.401
## - N3.m12  1   0.06236 5.9122 -58.348
## - N3.m11  1   0.13297 5.9828 -57.849
## - N3.m02  1   0.20345 6.0532 -57.357
## - N3.m03  1   0.22452 6.0743 -57.211
## - N3.m04  1   0.24472 6.0945 -57.072
## <none>                 5.8498 -56.793
## - N3.m01  1   0.30177 6.1516 -56.680
## - N3.m07  1   0.66253 6.5123 -54.287
## - N3.m08  1   0.76457 6.6144 -53.634
##
## Step:  AIC=-58.79
## Typhoons.sqrt ~ N3.m01 + N3.m02 + N3.m03 + N3.m04 + N3.m06 +
##      N3.m07 + N3.m08 + N3.m09 + N3.m10 + N3.m11 + N3.m12
##
##      Df Sum of Sq  RSS    AIC
## - N3.m09  1   0.00912 5.8593 -60.725
## - N3.m10  1   0.02524 5.8754 -60.609
## - N3.m12  1   0.06198 5.9122 -60.348
## - N3.m06  1   0.08090 5.9311 -60.213
```

```

## - N3.m11 1 0.13291 5.9831 -59.847
## - N3.m02 1 0.20317 6.0534 -59.356
## - N3.m03 1 0.23849 6.0887 -59.112
## <none> 5.8502 -58.790
## - N3.m01 1 0.30431 6.1545 -58.660
## - N3.m04 1 0.35735 6.2075 -58.300
## - N3.m07 1 0.68894 6.5391 -56.114
## - N3.m08 1 0.76450 6.6147 -55.632
##
## Step: AIC=-60.72
## Typhoons.sqrt ~ N3.m01 + N3.m02 + N3.m03 + N3.m04 + N3.m06 +
## N3.m07 + N3.m08 + N3.m10 + N3.m11 + N3.m12
##
## Df Sum of Sq RSS AIC
## - N3.m12 1 0.07135 5.9307 -62.216
## - N3.m10 1 0.07386 5.9332 -62.199
## - N3.m06 1 0.09558 5.9549 -62.045
## - N3.m11 1 0.15699 6.0163 -61.614
## - N3.m02 1 0.20350 6.0628 -61.291
## - N3.m03 1 0.22978 6.0891 -61.109
## <none> 5.8593 -60.725
## - N3.m01 1 0.30827 6.1676 -60.571
## - N3.m04 1 0.36345 6.2228 -60.197
## - N3.m07 1 0.70101 6.5603 -57.978
## - N3.m08 1 1.20206 7.0614 -54.887
##
## Step: AIC=-62.22
## Typhoons.sqrt ~ N3.m01 + N3.m02 + N3.m03 + N3.m04 + N3.m06 +
## N3.m07 + N3.m08 + N3.m10 + N3.m11
##
## Df Sum of Sq RSS AIC
## - N3.m06 1 0.04828 5.9789 -63.876
## - N3.m10 1 0.05955 5.9902 -63.797
## - N3.m11 1 0.08996 6.0206 -63.584
## - N3.m02 1 0.16840 6.0991 -63.040
## - N3.m03 1 0.23199 6.1627 -62.605
## - N3.m01 1 0.26210 6.1928 -62.400
## <none> 5.9307 -62.216
## - N3.m04 1 0.33635 6.2670 -61.899
## - N3.m07 1 0.65826 6.5889 -59.796
## - N3.m08 1 1.22330 7.1540 -56.340
##
## Step: AIC=-63.88
## Typhoons.sqrt ~ N3.m01 + N3.m02 + N3.m03 + N3.m04 + N3.m07 +
## N3.m08 + N3.m10 + N3.m11
##
## Df Sum of Sq RSS AIC
## - N3.m10 1 0.06059 6.0395 -65.452
## - N3.m11 1 0.11132 6.0903 -65.101
## - N3.m02 1 0.17510 6.1541 -64.663
## - N3.m01 1 0.24372 6.2227 -64.198
## - N3.m03 1 0.26404 6.2430 -64.061
## - N3.m04 1 0.28809 6.2670 -63.899
## <none> 5.9789 -63.876

```



```

## - N3.m07 1 0.80402 6.7830 -60.577
## - N3.m08 1 1.21638 7.1953 -58.098
##
## Step: AIC=-65.45
## Typhoons.sqrt ~ N3.m01 + N3.m02 + N3.m03 + N3.m04 + N3.m07 +
## N3.m08 + N3.m11
##
##      Df Sum of Sq  RSS    AIC
## - N3.m11 1 0.08021 6.1198 -66.898
## - N3.m02 1 0.16714 6.2067 -66.306
## - N3.m03 1 0.22274 6.2623 -65.931
## - N3.m04 1 0.26190 6.3014 -65.669
## - N3.m01 1 0.26445 6.3040 -65.652
## <none>      6.0395 -65.452
## - N3.m07 1 0.77156 6.8111 -62.403
## - N3.m08 1 1.17650 7.2160 -59.977
##
## Step: AIC=-66.9
## Typhoons.sqrt ~ N3.m01 + N3.m02 + N3.m03 + N3.m04 + N3.m07 +
## N3.m08
##
##      Df Sum of Sq  RSS    AIC
## - N3.m03 1 0.20631 6.3261 -67.506
## - N3.m02 1 0.21851 6.3383 -67.425
## - N3.m04 1 0.25485 6.3746 -67.185
## <none>      6.1198 -66.898
## - N3.m01 1 0.35359 6.4733 -66.539
## - N3.m07 1 0.72521 6.8450 -64.195
## - N3.m08 1 1.61744 7.7372 -59.048
##
## Step: AIC=-67.51
## Typhoons.sqrt ~ N3.m01 + N3.m02 + N3.m04 + N3.m07 + N3.m08
##
##      Df Sum of Sq  RSS    AIC
## - N3.m02 1 0.07950 6.4056 -68.981
## - N3.m04 1 0.11558 6.4416 -68.745
## <none>      6.3261 -67.506
## - N3.m01 1 0.33157 6.6576 -67.360
## - N3.m07 1 0.95327 7.2793 -63.610
## - N3.m08 1 2.04887 8.3749 -57.722
##
## Step: AIC=-68.98
## Typhoons.sqrt ~ N3.m01 + N3.m04 + N3.m07 + N3.m08
##
##      Df Sum of Sq  RSS    AIC
## <none>      6.4056 -68.981
## - N3.m04 1 0.41851 6.8241 -68.323
## - N3.m01 1 0.83446 7.2400 -65.838
## - N3.m07 1 1.10150 7.5071 -64.317
## - N3.m08 1 2.35255 8.7581 -57.843
##
## Call:
## lm(formula = Typhoons.sqrt ~ N3.m01 + N3.m04 + N3.m07 + N3.m08,

```

```
##      data = alldata[, -(1:2)])
##
## Coefficients:
## (Intercept)      N3.m01      N3.m04      N3.m07      N3.m08
##      4.1514      -0.1994      0.2667      0.7309     -0.9817
```

## Bidirectional selection (stepwise) by means of AIC

Stepwise selection (or sequential replacement), which is a combination of forward and backward selections. You start with no predictors, then sequentially add the most contributive predictors (like forward selection). After adding each new variable, remove any variables that no longer provide an improvement in the model fit (like backward selection).

```
fit.start<-lm(Typhoons.sqrt~N3.m05+N3.m08+N3.m12)
```

```
step(fit.start,scope=list(lower=fit.null,upper=fit.full),
      direction="both",k=k,trace=1)
```

```
## Start:  AIC=-61.8
## Typhoons.sqrt ~ N3.m05 + N3.m08 + N3.m12
##
##      Df Sum of Sq  RSS    AIC
## + N3.m07  1   0.82627 7.1446 -64.395
## - N3.m05  1   0.00386 7.9748 -63.778
## - N3.m12  1   0.02444 7.9954 -63.670
## + N3.m03  1   0.49271 7.4782 -62.478
## + N3.m01  1   0.39584 7.5751 -61.938
## + N3.m06  1   0.39011 7.5808 -61.906
## <none>                 7.9709 -61.799
## - N3.m08  1   0.44451 8.4154 -61.519
## + N3.m02  1   0.21139 7.7595 -60.927
## + N3.m09  1   0.17045 7.8005 -60.706
## + N3.m11  1   0.11741 7.8535 -60.422
## + N3.m10  1   0.02113 7.9498 -59.910
## + N3.m04  1   0.00460 7.9663 -59.823
##
## Step:  AIC=-64.39
## Typhoons.sqrt ~ N3.m05 + N3.m08 + N3.m12 + N3.m07
##
##      Df Sum of Sq  RSS    AIC
## - N3.m05  1   0.04058 7.1852 -66.157
## - N3.m12  1   0.06343 7.2081 -66.024
## + N3.m01  1   0.56143 6.5832 -65.832
## + N3.m03  1   0.46888 6.6758 -65.246
## + N3.m02  1   0.36256 6.7821 -64.582
## <none>                 7.1446 -64.395
## + N3.m11  1   0.05119 7.0935 -62.697
## + N3.m04  1   0.00574 7.1389 -62.429
## + N3.m06  1   0.00040 7.1442 -62.397
## + N3.m10  1   0.00025 7.1444 -62.396
## + N3.m09  1   0.00023 7.1444 -62.396
## - N3.m07  1   0.82627 7.9709 -61.799
## - N3.m08  1   1.26896 8.4136 -59.528
##
## Step:  AIC=-66.16
```

```

## Typhoons.sqrt ~ N3.m08 + N3.m12 + N3.m07
##
##      Df Sum of Sq  RSS    AIC
## - N3.m12  1   0.05979 7.2450 -67.809
## + N3.m01  1   0.50843 6.6768 -67.239
## + N3.m03  1   0.33983 6.8454 -66.192
## <none>                 7.1852 -66.157
## + N3.m02  1   0.32616 6.8591 -66.108
## + N3.m11  1   0.04575 7.1395 -64.425
## + N3.m05  1   0.04058 7.1446 -64.395
## + N3.m06  1   0.02217 7.1630 -64.287
## + N3.m04  1   0.01991 7.1653 -64.274
## + N3.m10  1   0.00044 7.1848 -64.160
## + N3.m09  1   0.00027 7.1850 -64.159
## - N3.m07  1   0.78956 7.9748 -63.778
## - N3.m08  1   1.27243 8.4576 -61.309
##
## Step:  AIC=-67.81
## Typhoons.sqrt ~ N3.m08 + N3.m07
##
##      Df Sum of Sq  RSS    AIC
## + N3.m01  1   0.42093 6.8241 -68.323
## <none>                 7.2450 -67.809
## + N3.m03  1   0.23821 7.0068 -67.213
## + N3.m02  1   0.23284 7.0122 -67.181
## + N3.m11  1   0.08857 7.1564 -66.326
## + N3.m12  1   0.05979 7.1852 -66.157
## + N3.m10  1   0.04186 7.2031 -66.052
## + N3.m05  1   0.03693 7.2081 -66.024
## + N3.m06  1   0.03684 7.2082 -66.023
## + N3.m09  1   0.03008 7.2149 -65.984
## + N3.m04  1   0.00499 7.2400 -65.838
## - N3.m07  1   0.75435 7.9994 -65.649
## - N3.m08  1   1.64441 8.8894 -61.218
##
## Step:  AIC=-68.32
## Typhoons.sqrt ~ N3.m08 + N3.m07 + N3.m01
##
##      Df Sum of Sq  RSS    AIC
## + N3.m04  1   0.41851 6.4056 -68.981
## + N3.m02  1   0.38243 6.4416 -68.745
## <none>                 6.8241 -68.323
## - N3.m01  1   0.42093 7.2450 -67.809
## + N3.m11  1   0.18293 6.6411 -67.464
## + N3.m12  1   0.14729 6.6768 -67.239
## + N3.m10  1   0.11298 6.7111 -67.024
## + N3.m05  1   0.06803 6.7560 -66.744
## + N3.m03  1   0.06011 6.7640 -66.695
## + N3.m09  1   0.02278 6.8013 -66.463
## + N3.m06  1   0.00027 6.8238 -66.325
## - N3.m07  1   1.01118 7.8353 -64.520
## - N3.m08  1   1.98286 8.8069 -59.609
##
## Step:  AIC=-68.98

```

```
## Typhoons.sqrt ~ N3.m08 + N3.m07 + N3.m01 + N3.m04
##
##           Df Sum of Sq    RSS    AIC
## <none>                6.4056 -68.981
## - N3.m04    1    0.41851 6.8241 -68.323
## + N3.m06    1    0.13094 6.2746 -67.849
## + N3.m05    1    0.12706 6.2785 -67.823
## + N3.m11    1    0.09771 6.3079 -67.627
## + N3.m02    1    0.07950 6.3261 -67.506
## + N3.m12    1    0.07444 6.3311 -67.472
## + N3.m03    1    0.06729 6.3383 -67.425
## + N3.m10    1    0.05422 6.3513 -67.338
## + N3.m09    1    0.00994 6.3956 -67.046
## - N3.m01    1    0.83446 7.2400 -65.838
## - N3.m07    1    1.10150 7.5071 -64.317
## - N3.m08    1    2.35255 8.7581 -57.843

##
## Call:
## lm(formula = Typhoons.sqrt ~ N3.m08 + N3.m07 + N3.m01 + N3.m04)
##
## Coefficients:
## (Intercept)      N3.m08      N3.m07      N3.m01      N3.m04
##      4.1514      -0.9817       0.7309      -0.1994       0.2667
```

## Bidirectional selection (stepwise) by means of AIC

```
k <- 2
```

```
step.aic<-step(fit.start,scope=list(lower=fit.null,upper=fit.full),
               direction="both",k=k,trace=1)
```

```
## Start:  AIC=-61.8
## Typhoons.sqrt ~ N3.m05 + N3.m08 + N3.m12
##
##           Df Sum of Sq    RSS    AIC
## + N3.m07    1    0.82627 7.1446 -64.395
## - N3.m05    1    0.00386 7.9748 -63.778
## - N3.m12    1    0.02444 7.9954 -63.670
## + N3.m03    1    0.49271 7.4782 -62.478
## + N3.m01    1    0.39584 7.5751 -61.938
## + N3.m06    1    0.39011 7.5808 -61.906
## <none>                7.9709 -61.799
## - N3.m08    1    0.44451 8.4154 -61.519
## + N3.m02    1    0.21139 7.7595 -60.927
## + N3.m09    1    0.17045 7.8005 -60.706
## + N3.m11    1    0.11741 7.8535 -60.422
## + N3.m10    1    0.02113 7.9498 -59.910
## + N3.m04    1    0.00460 7.9663 -59.823
##
## Step:  AIC=-64.39
## Typhoons.sqrt ~ N3.m05 + N3.m08 + N3.m12 + N3.m07
##
##           Df Sum of Sq    RSS    AIC
## - N3.m05    1    0.04058 7.1852 -66.157
```

```

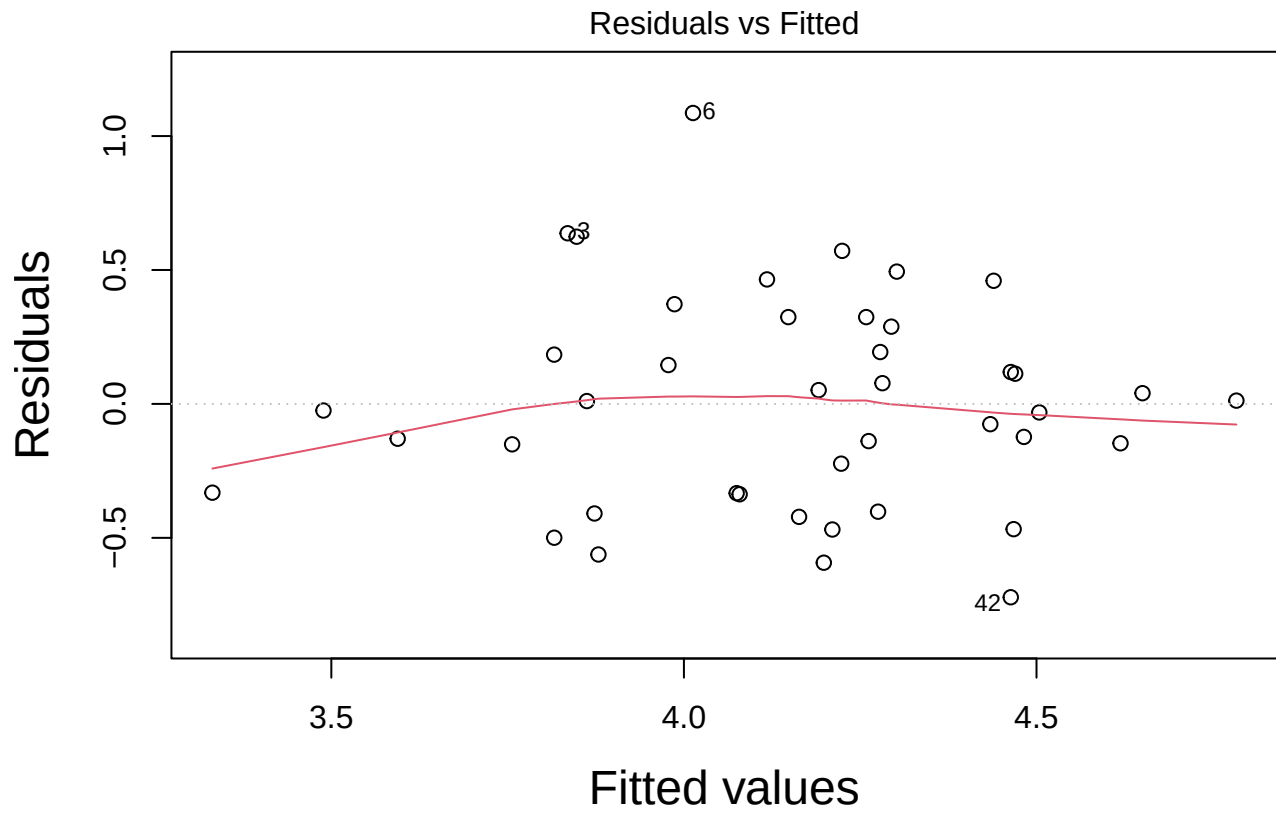
## - N3.m12 1 0.06343 7.2081 -66.024
## + N3.m01 1 0.56143 6.5832 -65.832
## + N3.m03 1 0.46888 6.6758 -65.246
## + N3.m02 1 0.36256 6.7821 -64.582
## <none> 7.1446 -64.395
## + N3.m11 1 0.05119 7.0935 -62.697
## + N3.m04 1 0.00574 7.1389 -62.429
## + N3.m06 1 0.00040 7.1442 -62.397
## + N3.m10 1 0.00025 7.1444 -62.396
## + N3.m09 1 0.00023 7.1444 -62.396
## - N3.m07 1 0.82627 7.9709 -61.799
## - N3.m08 1 1.26896 8.4136 -59.528
##
## Step: AIC=-66.16
## Typhoons.sqrt ~ N3.m08 + N3.m12 + N3.m07
##
## Df Sum of Sq RSS AIC
## - N3.m12 1 0.05979 7.2450 -67.809
## + N3.m01 1 0.50843 6.6768 -67.239
## + N3.m03 1 0.33983 6.8454 -66.192
## <none> 7.1852 -66.157
## + N3.m02 1 0.32616 6.8591 -66.108
## + N3.m11 1 0.04575 7.1395 -64.425
## + N3.m05 1 0.04058 7.1446 -64.395
## + N3.m06 1 0.02217 7.1630 -64.287
## + N3.m04 1 0.01991 7.1653 -64.274
## + N3.m10 1 0.00044 7.1848 -64.160
## + N3.m09 1 0.00027 7.1850 -64.159
## - N3.m07 1 0.78956 7.9748 -63.778
## - N3.m08 1 1.27243 8.4576 -61.309
##
## Step: AIC=-67.81
## Typhoons.sqrt ~ N3.m08 + N3.m07
##
## Df Sum of Sq RSS AIC
## + N3.m01 1 0.42093 6.8241 -68.323
## <none> 7.2450 -67.809
## + N3.m03 1 0.23821 7.0068 -67.213
## + N3.m02 1 0.23284 7.0122 -67.181
## + N3.m11 1 0.08857 7.1564 -66.326
## + N3.m12 1 0.05979 7.1852 -66.157
## + N3.m10 1 0.04186 7.2031 -66.052
## + N3.m05 1 0.03693 7.2081 -66.024
## + N3.m06 1 0.03684 7.2082 -66.023
## + N3.m09 1 0.03008 7.2149 -65.984
## + N3.m04 1 0.00499 7.2400 -65.838
## - N3.m07 1 0.75435 7.9994 -65.649
## - N3.m08 1 1.64441 8.8894 -61.218
##
## Step: AIC=-68.32
## Typhoons.sqrt ~ N3.m08 + N3.m07 + N3.m01
##
## Df Sum of Sq RSS AIC
## + N3.m04 1 0.41851 6.4056 -68.981

```

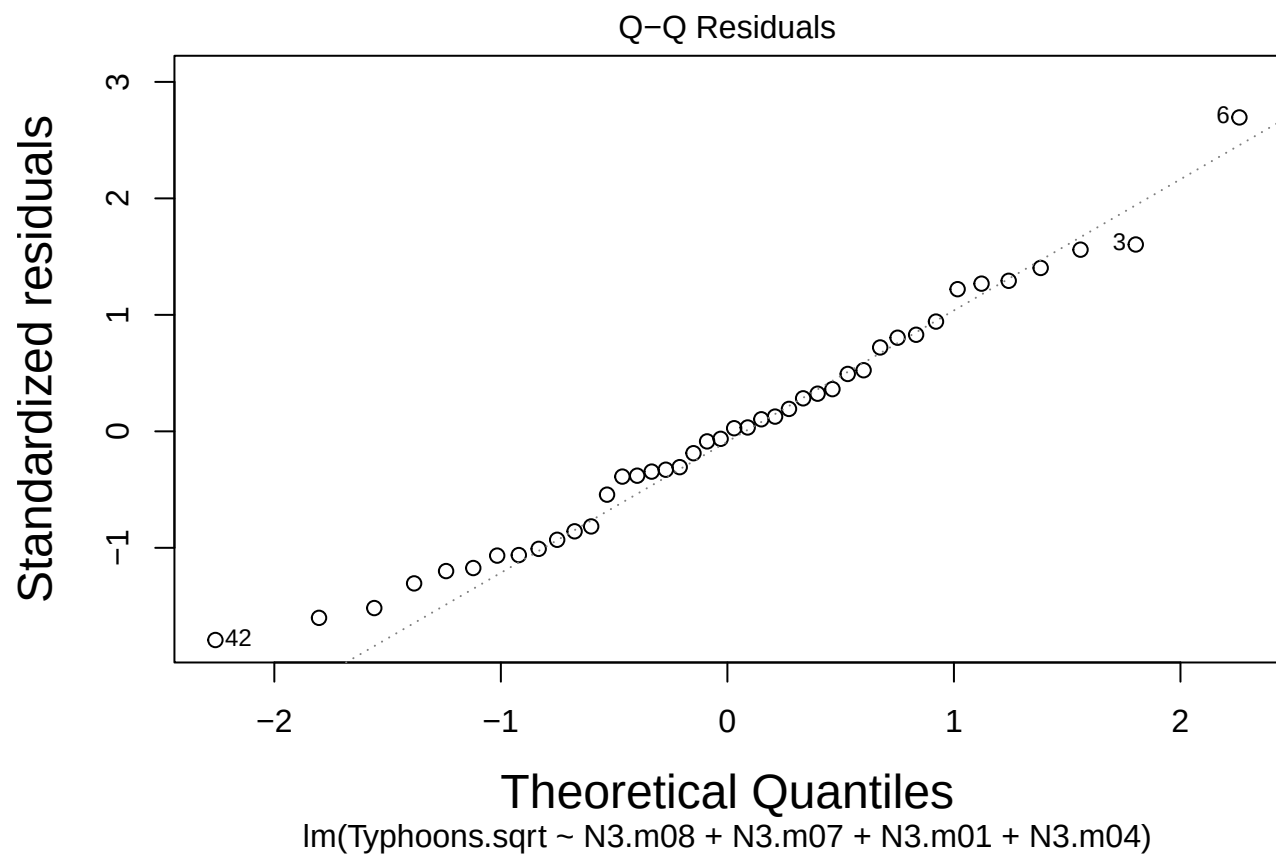
```
## + N3.m02 1 0.38243 6.4416 -68.745
## <none> 6.8241 -68.323
## - N3.m01 1 0.42093 7.2450 -67.809
## + N3.m11 1 0.18293 6.6411 -67.464
## + N3.m12 1 0.14729 6.6768 -67.239
## + N3.m10 1 0.11298 6.7111 -67.024
## + N3.m05 1 0.06803 6.7560 -66.744
## + N3.m03 1 0.06011 6.7640 -66.695
## + N3.m09 1 0.02278 6.8013 -66.463
## + N3.m06 1 0.00027 6.8238 -66.325
## - N3.m07 1 1.01118 7.8353 -64.520
## - N3.m08 1 1.98286 8.8069 -59.609
##
## Step: AIC=-68.98
## Typhoons.sqrt ~ N3.m08 + N3.m07 + N3.m01 + N3.m04
##
## Df Sum of Sq RSS AIC
## <none> 6.4056 -68.981
## - N3.m04 1 0.41851 6.8241 -68.323
## + N3.m06 1 0.13094 6.2746 -67.849
## + N3.m05 1 0.12706 6.2785 -67.823
## + N3.m11 1 0.09771 6.3079 -67.627
## + N3.m02 1 0.07950 6.3261 -67.506
## + N3.m12 1 0.07444 6.3311 -67.472
## + N3.m03 1 0.06729 6.3383 -67.425
## + N3.m10 1 0.05422 6.3513 -67.338
## + N3.m09 1 0.00994 6.3956 -67.046
## - N3.m01 1 0.83446 7.2400 -65.838
## - N3.m07 1 1.10150 7.5071 -64.317
## - N3.m08 1 2.35255 8.7581 -57.843
```

Analysis of residuals

```
plot(step.aic,which = 1,sub.caption = "",cex.lab=1.5)
```

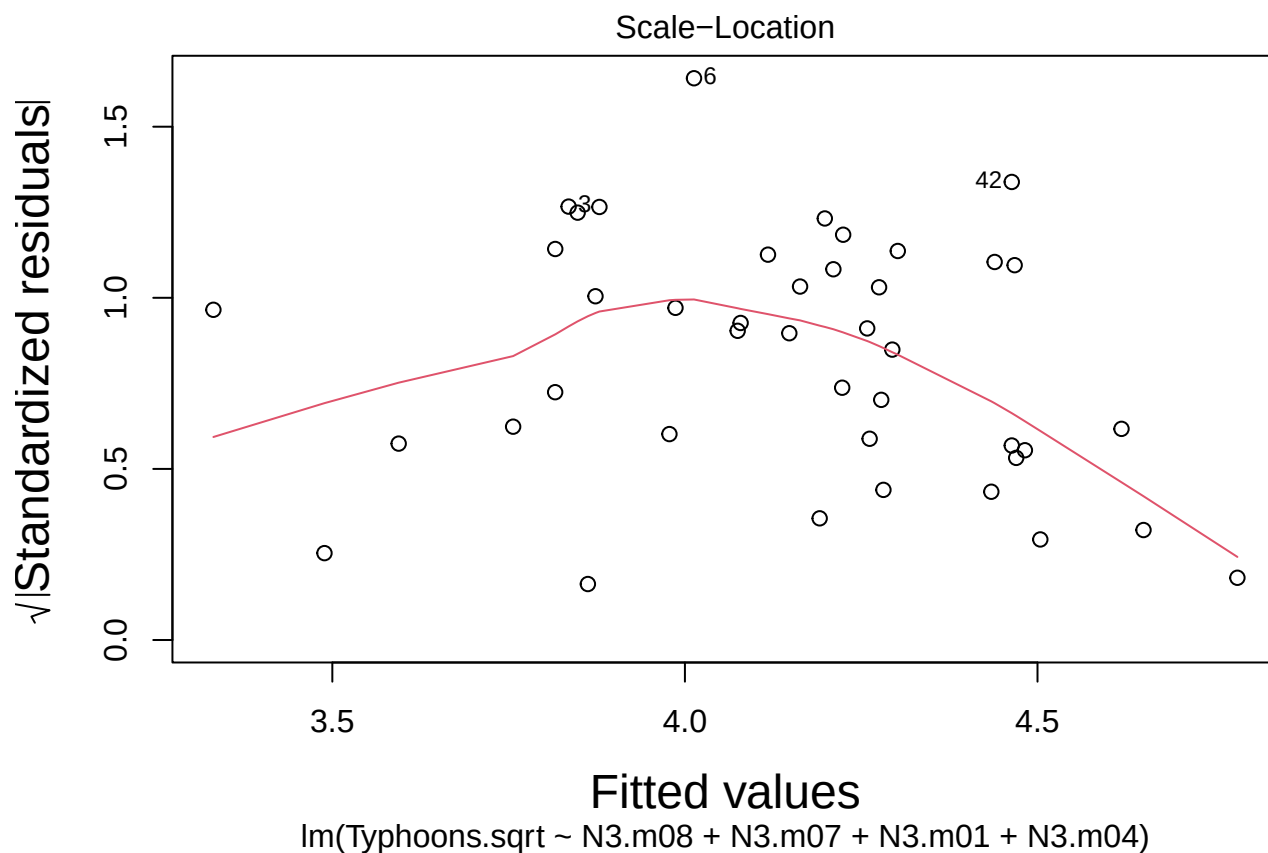


```
plot(step.aic,which = 2,cex.lab=1.5)
```



```
plot(step.aic, which = 3, cex.lab=1.5)
```





Bidirectional selection (stepwise) by means of BIC

```
k <- log(length(Typhoons.sqrt))
```

More penalty !

```
step.bic <- step(fit.start, scope=list(lower=fit.null, upper=fit.full),
                 direction="both", k=k, trace=1)
```

```
## Start:  AIC=-54.85
## Typhoons.sqrt ~ N3.m05 + N3.m08 + N3.m12
##
##      Df Sum of Sq  RSS   AIC
## - N3.m05  1   0.00386 7.9748 -58.565
## - N3.m12  1   0.02444 7.9954 -58.457
## - N3.m08  1   0.44451 8.4154 -56.306
## + N3.m07  1   0.82627 7.1446 -55.707
## <none>                 7.9709 -54.848
## + N3.m03  1   0.49271 7.4782 -53.790
## + N3.m01  1   0.39584 7.5751 -53.249
## + N3.m06  1   0.39011 7.5808 -53.218
## + N3.m02  1   0.21139 7.7595 -52.239
## + N3.m09  1   0.17045 7.8005 -52.018
## + N3.m11  1   0.11741 7.8535 -51.733
## + N3.m10  1   0.02113 7.9498 -51.222
## + N3.m04  1   0.00460 7.9663 -51.134
##
```

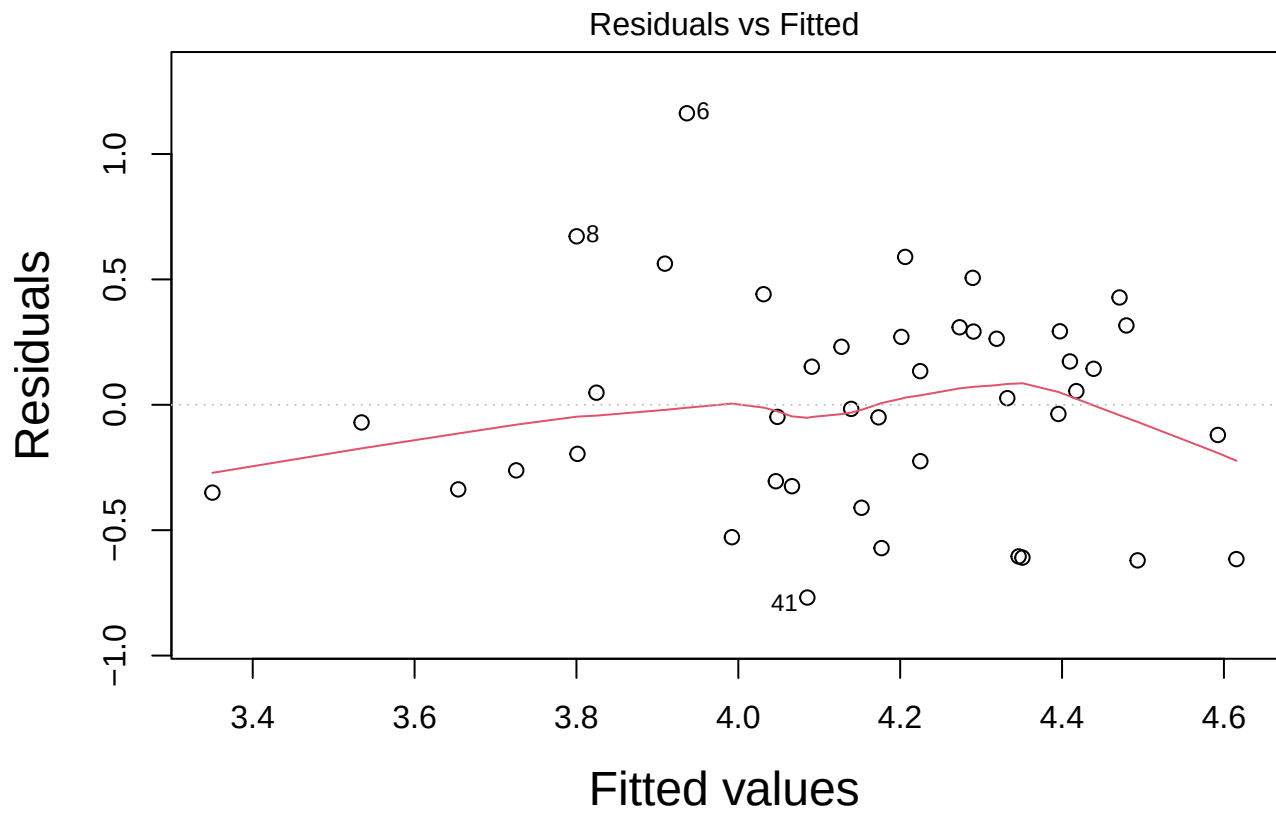
```

## Step: AIC=-58.57
## Typhoons.sqrt ~ N3.m08 + N3.m12
##
##      Df Sum of Sq    RSS    AIC
## - N3.m12  1   0.02458  7.9994 -62.174
## - N3.m08  1   0.51491  8.4897 -59.675
## + N3.m07  1   0.78956  7.1852 -59.206
## <none>                 7.9748 -58.565
## + N3.m06  1   0.19580  7.7790 -55.872
## + N3.m01  1   0.19179  7.7830 -55.850
## + N3.m09  1   0.17025  7.8045 -55.734
## + N3.m03  1   0.14132  7.8335 -55.579
## + N3.m11  1   0.12051  7.8543 -55.467
## + N3.m02  1   0.07347  7.9013 -55.216
## + N3.m10  1   0.02245  7.9523 -54.946
## + N3.m05  1   0.00386  7.9709 -54.848
## + N3.m04  1   0.00051  7.9743 -54.830
##
## Step: AIC=-62.17
## Typhoons.sqrt ~ N3.m08
##
##      Df Sum of Sq    RSS    AIC
## + N3.m07  1   0.75435  7.2450 -62.596
## <none>                 7.9994 -62.174
## + N3.m09  1   0.18038  7.8190 -59.394
## + N3.m01  1   0.16410  7.8353 -59.307
## + N3.m06  1   0.16038  7.8390 -59.287
## + N3.m03  1   0.10333  7.8960 -58.982
## + N3.m11  1   0.06505  7.9343 -58.779
## + N3.m02  1   0.05142  7.9479 -58.707
## + N3.m10  1   0.04408  7.9553 -58.668
## + N3.m12  1   0.02458  7.9748 -58.565
## + N3.m05  1   0.00400  7.9954 -58.457
## + N3.m04  1   0.00369  7.9957 -58.455
## - N3.m08  1   2.51842 10.5178 -54.416
##
## Step: AIC=-62.6
## Typhoons.sqrt ~ N3.m08 + N3.m07
##
##      Df Sum of Sq    RSS    AIC
## <none>                 7.2450 -62.596
## - N3.m07  1   0.75435  7.9994 -62.174
## + N3.m01  1   0.42093  6.8241 -61.372
## + N3.m03  1   0.23821  7.0068 -60.262
## + N3.m02  1   0.23284  7.0122 -60.230
## + N3.m11  1   0.08857  7.1564 -59.375
## + N3.m12  1   0.05979  7.1852 -59.206
## + N3.m10  1   0.04186  7.2031 -59.102
## + N3.m05  1   0.03693  7.2081 -59.073
## + N3.m06  1   0.03684  7.2082 -59.072
## + N3.m09  1   0.03008  7.2149 -59.033
## + N3.m04  1   0.00499  7.2400 -58.887
## - N3.m08  1   1.64441  8.8894 -57.743

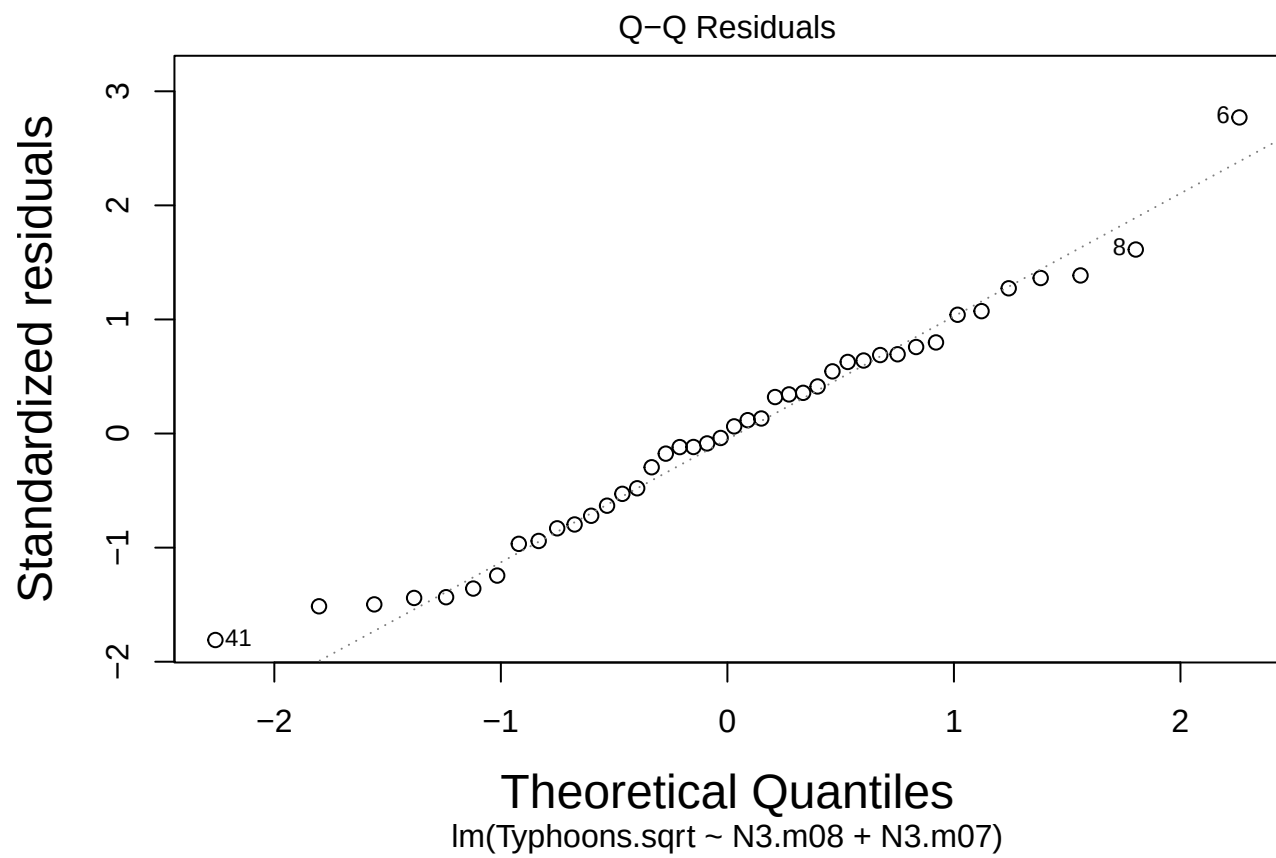
```

Analysis of residuals

```
plot(step.bic, which = 1, sub.caption = "", cex.lab=1.5)
```



```
plot(step.bic, which = 2, cex.lab=1.5)
```



```
plot(step.bic, which = 3, cex.lab=1.5)
```

